

| Q7        | Suggested Solutions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Mark                                               |
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| 7(a)(i)   | <p>For multiple-source interference, with source separation <math>d</math> emitting waves of wavelength <math>\lambda</math>, maxima are formed at angles <math>\theta</math> to the horizontal:</p> $d \sin \theta = m\lambda = m \frac{c}{f}, \text{ where } m = 0, 1, 2, \dots$ $6.0 \sin \theta = m \left( \frac{3.00 \times 10^8}{30.0 \times 10^6} \right)$ $\sin \theta = m \times \frac{10}{6}$ <p>But since <math>\sin \theta \leq 1</math>, this equation is only valid if the order <math>m = 0</math>.<br/>OR<br/>Solving for <math>m</math> at <math>\theta = 90^\circ</math>, we get <math>m = \frac{3}{5} = 0.60 &lt; 1</math></p> <p>Therefore, there is only one central maximum (the zeroth order maximum)</p> | <p><b>M1</b></p> <p><b>M1</b></p> <p><b>A0</b></p> |
| 7(a)(ii)  | $\sin \frac{\alpha}{2} = \frac{\lambda}{b}$ $= \frac{c}{fb}$ $\frac{\alpha}{2} = \sin^{-1} \left( \frac{3.00 \times 10^8}{(30.0 \times 10^6)(42.0)} \right)$ $= 13.77^\circ$ $\alpha = 13.77^\circ \times 2$ $= 27.5^\circ$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p><b>M1</b></p> <p><b>M1</b></p> <p><b>A1</b></p> |
| 7(b)(i)   | From Fig 7.2, at 400 km from station, altitude = 5400 m                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>A1</b>                                          |
| 7(b)(ii)  | $\text{speed} = \frac{\text{distance travelled}}{\text{time}} = \frac{AB}{\tau} = \frac{5000}{51.4} = 97.3 \text{ ms}^{-1}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>A1</b>                                          |
| 7(b)(iii) | <p>From (ii), speed <math>u_x = 97.3 \text{ ms}^{-1}</math>.</p> $s_y = \frac{1}{2}gt^2 \text{ ----- (1)}$ $s_x = u_x t \text{ ----- (2)}$ $\frac{(2)^2}{(1)} : \frac{s_x^2}{s_y} = \frac{2u_x^2}{g}$ $s_x = \sqrt{\frac{2u_x^2 s_y}{g}} = \sqrt{\frac{2(97.3)^2 (5400)}{9.81}}$ $= 3230 \text{ m}$ <p>OR</p>                                                                                                                                                                                                                                                                                                                                                                                                                    | <p><b>M1</b></p> <p><b>M1</b></p> <p><b>A1</b></p> |



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| <b>7(c)(iii)</b> | <p>Net force = 0, thus drag force <math>F_D</math> = thrust <math>T</math></p> <p><math>P = 2 \times 990 \times 10^3 = 1.98 \times 10^6 \text{ W}</math></p> <p>Power <math>P</math> = Thrust <math>T</math> x speed <math>v</math></p> $P = Tv \Rightarrow T = \frac{P}{v} = \frac{1.98 \times 10^6 \text{ W}}{200 \times \frac{1000}{3600} \text{ ms}^{-1}}$ $= 3.56 \times 10^4 \text{ N}$ <p>Therefore <math>F_D = T = 3.56 \times 10^4 \text{ N}</math></p> | <p><b>B1</b></p> <p><b>B1</b></p> <p><b>M1</b></p> <p><b>A1</b></p> |
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